

Pandas Tutorial 4: Selecting Rows and Filtering Data

In this tutorial, students will learn how to select rows and columns and how to filter data using conditions. These operations are very important in Exploratory Data Analysis.

1. Selecting a Single Column

To select one column from a DataFrame:

```
df["ColumnName"]
```

Example:

```
df["Age"]
```

This returns a Series containing only the selected column.

2. Selecting Multiple Columns

To select multiple columns, use double brackets:

```
df[["Name", "Age"]]
```

This returns a new DataFrame with selected columns.

3. Selecting Rows by Index Using loc

The loc method is used to select rows using labels or index names.

```
df.loc[0]
```

This selects the row with index 0.

To select multiple rows:

```
df.loc[0:3]
```

To select specific rows and columns:

```
df.loc[0:2, ["Name", "Marks"]]
```

loc is label based indexing.

4. Selecting Rows by Position Using iloc

The iloc method is used for position based indexing.

```
df.iloc[0]
```

This selects the first row.

To select multiple rows:

```
df.iloc[0:3]
```

To select rows and columns using position:

```
df.iloc[0:3, 0:2]
```

iloc works using numerical index positions.

5. Filtering Data Using Conditions

Filtering is used to select rows based on conditions.

Example:

```
df[df["Age"] > 21]
```

This returns rows where Age is greater than 21.

6. Multiple Conditions

To apply multiple conditions, use logical operators.

AND condition:

```
df[(df["Age"] > 20) & (df["Marks"] > 85)]
```

OR condition:

```
df[(df["Age"] > 22) | (df["Marks"] > 90)]
```

Important:

- Use & for AND.
- Use | for OR.
- Always wrap each condition inside brackets.

7. Using isin Method

The isin method is used to filter categorical values.

Example:

```
df[df["Name"].isin(["Ali", "Sara"])]
```

This returns rows where Name is either Ali or Sara.

8. Sorting Data

To sort data by a column:

```
df.sort_values("Marks")
```

To sort in descending order:

```
df.sort_values("Marks", ascending=False)
```

Sorting helps in understanding highest and lowest values in dataset.

9. Resetting Index

After filtering or sorting, index may change.

To reset index:

```
df.reset_index(drop=True)
```

This removes old index and creates a new clean index.

10. Conclusion of Tutorial 3

In this tutorial, students have learned:

1. How to select single and multiple columns.
2. How to use loc and iloc for row selection.
3. How to filter data using conditions.
4. How to apply multiple conditions.
5. How to sort data.
6. How to reset index.

These operations are fundamental for performing real world Exploratory Data Analysis.

In the next tutorial, we will learn about grouping data and aggregation functions.

This completes Pandas Tutorial 3.